

Exercise of Supervised Learning: Advanced Risk Minimization Part 1

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April 23, 2026

Exercise 1: Risk Minimizers for Generalized L2-Loss

Consider the regression learning setting, i.e., $\mathcal{Y} = \mathbb{R}$, and assume that your loss function of interest is $L(y, f(\mathbf{x})) = (m(y) - m(f(\mathbf{x})))^2$, where: $m : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous strictly monotone function.

Disclaimer: In the following we always assume that $\text{Var}(m(Y))$ exists.

(a) Consider the hypothesis space of a constant models

$\mathcal{H} = \{f : \mathcal{X} \rightarrow \mathbb{R} \mid f(\mathbf{x}) = \theta \ \forall \mathbf{x} \in \mathcal{X}\}$, where $\mathcal{X}$ is the feature space. Show that

$$\hat{f}(\mathbf{x}) = m^{-1} \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)$$

is the optimal constant model for the loss function above, where m^{-1} is the inverse function of m .

Solution to Question (a)

1. f is a **constant model**: $f(\mathbf{x}) = \theta$ for all $\mathbf{x}$.
2. The empirical risk can be formulated as:

$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^n \left(m(y^{(i)}) - m(f(\mathbf{x}^{(i)})) \right)^2 = \sum_{i=1}^n \left(m(y^{(i)}) - m(\theta) \right)^2.$$

3. $\mathcal{R}_{\text{emp}}(f)$ is **strictly convex** (because MSE loss and m is strictly monotone). So the minimum is unique, and can be computed by solving $\partial \mathcal{R}_{\text{emp}}(f) / \partial \theta = \mathbf{0}$.

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Solution to Question (a): Continued

Goal: Compute the optimal θ by solving $\partial \mathcal{R}_{\text{emp}}(f)/\partial \theta = \mathbf{0}$.

1. Compute the derivative:

$$\frac{\partial \mathcal{R}_{\text{emp}}(f)}{\partial \theta} = 2 \sum_{i=1}^n (m(y^{(i)}) - m(\theta)) \cdot \frac{\partial m(\theta)}{\partial \theta} = 0$$

2. Using the fact that $\frac{\partial m(\theta)}{\partial \theta}$ is constant for all i , we obtain:

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$$\begin{aligned} \sum_{i=1}^n (m(y^{(i)}) - m(\theta)) &= 0 \\ \Rightarrow \sum_{i=1}^n m(y^{(i)}) &= \sum_{i=1}^n m(\theta) \end{aligned}$$

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$$\sum_{i=1}^n (m(y^{(i)}) - m(\theta)) = 0$$

$$\Rightarrow \sum_{i=1}^n m(y^{(i)}) = \sum_{i=1}^n m(\theta)$$

$$\Rightarrow m(\theta) = \frac{1}{n} \sum_{i=1}^n m(y^{(i)})$$

$$\Rightarrow \theta^* = m^{-1} \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \quad \triangleleft$$

Question (b)

(b) Verify that the risk of the optimal constant model is $\mathcal{R}_L(\hat{f}) = (1 + \frac{1}{n})\text{Var}(m(y))$.

Recall that the risk of $\hat{f}$ is defined as

$$\mathcal{R}_L(\hat{f}) = \mathbb{E}_{xy}[L(y, \hat{f}(\mathbf{x}))]$$

Solution to Question (b)

$$\begin{aligned}\mathcal{R}_L(\hat{f}) &= \mathbb{E}_{xy}[L(y, \hat{f}(\mathbf{x}))] \\ &= \mathbb{E}_{xy}[(m(y) - m(\hat{f}(\mathbf{x})))^2]\end{aligned}$$

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Solution to Question (b): Continued

Now take a look at the second term: $-2 \cdot \mathbb{E}_{xy} \left[m(y) \cdot \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right]$.

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Why $\mathbb{E}_{xy} \left[\sum_{i=1}^n m(y^{(i)}) \right] = n \cdot \mathbb{E}_{xy}[m(y)]$?

- Linearity: pull $\mathbb{E}_{xy}$ inside the sum.
- (i) just labels different draws; all $y^{(i)}$ share y 's distribution, so $\mathbb{E}_{xy}[m(y^{(i)})] = \mathbb{E}_{xy}[m(y)]$.
- Sum of n equal terms $= n \cdot \mathbb{E}_{xy}[m(y)]$.

Analogy: expected sum of n dice rolls is $n \times 3.5$.

Solution to Question (b): Continued

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Solution (b): third term $\mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)^2 \right]$

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Example ($n = 3$). Let $m_i := m(y^{(i)})$.

$$\left(\frac{m_1 + m_2 + m_3}{3} \right)^2 = \frac{1}{9} \sum_{i,j=1}^3 m_i m_j$$

Expand all 9 pairs (i, j) :

$$\begin{array}{ccc} m_1^2 & m_1 m_2 & m_1 m_3 \\ m_2 m_1 & m_2^2 & m_2 m_3 \\ m_3 m_1 & m_3 m_2 & m_3^2 \end{array}$$

Group by color:

- ▶ **Diagonal** ($i = j$, n terms) $\rightsquigarrow \sum_i m_i^2$
- ▶ **Off-diag.** ($i \neq j$, $n(n-1)$ terms)
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$$= \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})^2] + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy} [m(y^{(i)}) m(y^{(j)})] \right)$$

▷ Split into **diagonal** ($i = j$, squared) and **off-diagonal** ($i \neq j$, cross) terms.

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Solution (b): third term, cont.

Continuing from the grouping form:

$$\begin{aligned} & \frac{1}{n^2} \left(\sum_{i=1}^n \mathbb{E}_{xy} [m(y^{(i)})^2] \right. \\ & \quad \left. + \sum_{i=1}^n \sum_{j \neq i} \mathbb{E}_{xy} [m(y^{(i)}) m(y^{(j)})] \right) \end{aligned}$$

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Cross term: independence trick.

For $i \neq j$, $m(y^{(i)})$ and $m(y^{(j)})$ are *independent* RVs, so

$$\mathbb{E}_{xy} [m(y^{(i)}) m(y^{(j)})] = \mathbb{E}_{xy} [m(y^{(i)})] \mathbb{E}_{xy} [m(y^{(j)})].$$

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Collapse by i.i.d.

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Combining the results so far, we get

$$\mathcal{R}_L(\hat{f}) = \mathbb{E}_{xy}[m(y)^2] - 2 \cdot \mathbb{E}_{xy} \left[m(y) \cdot \frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right] + \mathbb{E}_{xy} \left[\left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right) \right]$$

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Why does $\mathbb{E}_x$ vanish?

- ▶ Constant model $\Rightarrow$ integrands depend on y only.
- ▶ Marginalize: $\int p(x, y) dx = p(y)$.
- ▶ Hence $\mathbb{E}_{xy}[m(y)] = \mathbb{E}_y[m(y)]$; $\text{Var}(m(y))$ is over y .

Question (c)

Derive that the risk minimizer f^* is given by $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y)|\mathbf{x}])$.

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Q1: Which hypothesis space?

What $\mathcal{H}$ should we consider when deriving f^* ?

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The **unrestricted** space of all functions: $\mathcal{H} = \{f : \mathcal{X} \rightarrow \mathbb{R}\}$.

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Q2: What does “unrestricted” imply? Why?

What freedom does this give, and why does it matter?

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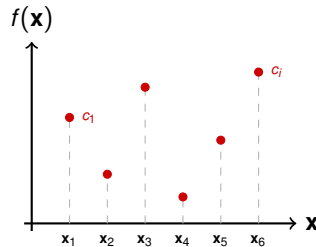
Q2: What does “unrestricted” imply? Why?

What freedom does this give, and why does it matter?

A2

For **each** $\mathbf{x}$, $f(\mathbf{x})$ can take any $c \in \mathbb{R}$ *independently*. $\rightsquigarrow$

Point-wise: minimize risk separately at every $\mathbf{x}$.



Each $\mathbf{x}_i$ gets its own optimal c_i

— no shared functional form.

Solution to Question (c)

Recall (c): point-wise optimization $\Rightarrow$ decompose

$$\mathbb{E}_{xy}[\cdot] = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[\cdot | \mathbf{x}]].$$

Apply the law of total expectation to the risk:

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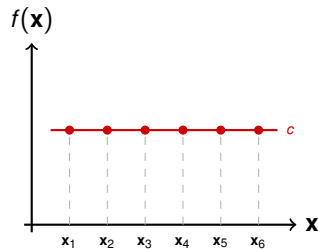
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Q2

Empirical or theoretical risk?

A2

The **theoretical** risk $\mathcal{R}_L(f) = \mathbb{E}_{xy}[L(y, f(\mathbf{x}))]$ (not empirical as in (a)).



One c for all $\mathbf{x}$.

Solution to Question (d)

The (theoretical) risk for a constant model $\bar{f}(\mathbf{x}) = c$ is:

$$\begin{aligned}\mathcal{R}_L(\bar{f}) &= \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [(m(y) - m(\bar{f}(\mathbf{x})))^2]] \\&= \int_y \int_{\mathbf{x}} (m(y) - m(\bar{f}(\mathbf{x})))^2 p(\mathbf{x}, y) d\mathbf{x} dy \\&= \int_y \int_{\mathbf{x}} (m(y) - m(c))^2 p(\mathbf{x}, y) d\mathbf{x} dy \\&= \int_y (m(y) - m(c))^2 p(y) dy \\&= \mathbb{E}_y [(m(y) - m(c))^2]\end{aligned}$$

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Therefore, the optimal constant model is

$$\bar{f}(\mathbf{x}) = c = m^{-1}(\mathbb{E}_y[m(y)])$$

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Solution to Question (d): Continued

The **risk** of the optimal constant model $\bar{f}(\mathbf{x}) = c = m^{-1}(\mathbb{E}_y[m(y)])$:

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The **risk** of the optimal constant model $\bar{f}(\mathbf{x}) = c = m^{-1}(\mathbb{E}_y[m(y)])$:

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Question (e)

(e): Recall the decomposition of the Bayes regret into the estimation and the approximation error. Show that the former is $\frac{1}{n} \text{Var}(m(y))$, while the latter is $\text{Var}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])$ for the optimal constant model $\hat{f}(\mathbf{x})$ if $\mathcal{H}$ consists of the constant models.

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Q1: What is the Bayes regret, and what are its two errors?

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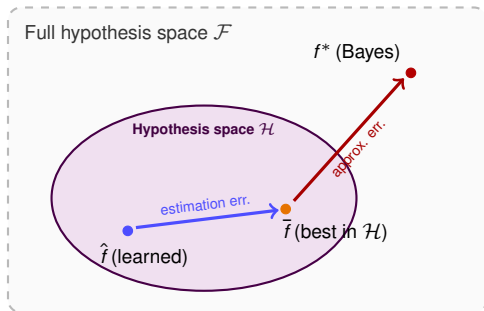
A1: Bayes regret = $\mathcal{R}_L(\hat{f}) - \mathcal{R}_L^*$, splits as:

Estimation error:

$$\mathcal{R}_L(\hat{f}) - \inf_{f \in \mathcal{H}} \mathcal{R}_L(f)$$

Approximation error:

$$\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*$$



Solution to Question (e): Roadmap

Three risk minimizers from earlier parts, connected by the two errors:

From (a)

$$\hat{f}(\mathbf{x}) = m^{-1} \left(\frac{1}{n} \sum_{i=1}^n m(y^{(i)}) \right)$$

$$\mathcal{R}_L(\hat{f}) = (1 + \frac{1}{n}) \text{Var}(m(y))$$

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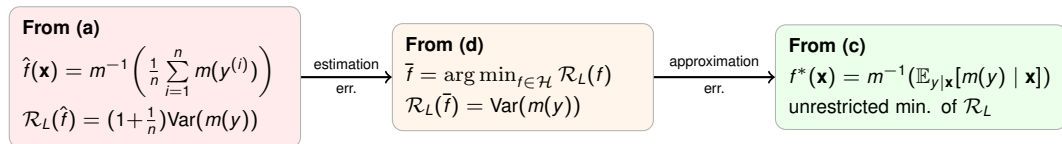
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$$\bar{f} = \arg \min_{f \in \mathcal{H}} \mathcal{R}_L(f)$$

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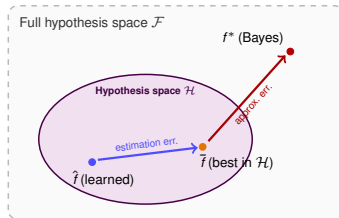
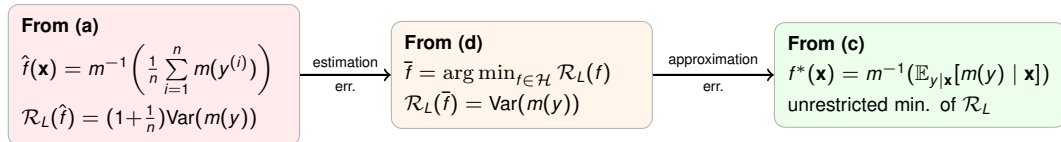
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Solution (e): Part 1 of 2 — Estimation Error

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From (c): $f^*(\mathbf{x}) = m^{-1}(\mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}])$,
so $m(f^*(\mathbf{x})) = \mathbb{E}_{y|\mathbf{x}}[m(y) \mid \mathbf{x}]$.

Law of total variance: $\text{Var}(Y) = \mathbb{E}_X [\text{Var}(Y \mid X)] + \text{Var}(\mathbb{E}[Y \mid X])$.